

Scalability & Future Plan

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Predictor
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1.	Local SQLite Database Constraints	Single-file database limits concurrent read/write operations when multiple users access the system simultaneously.	High
2.	API Rate Limits & Latency	Synchronous calls to the Google Gemini API can introduce slight latency during peak quiz generation requests.	Medium
3.	Single-User Local Session State	Lack of complete multi-tenant user authentication (OAuth/JWT) limits personalized user account management.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1.	User Load	Local Streamlit runtime suitable for single-user demonstration.	Migrate to microservices architecture deployed on Kubernetes or AWS ECS with automated load balancing.
2.	Data Storage	Local embedded SQLite database file (<code>knowledge_retention.db</code>).	Transition to a managed cloud relational database like PostgreSQL (AWS RDS / Supabase) for high availability and concurrent scaling.
3.	Performance	Direct API calls for quiz generation on each request.	Implement Redis caching for generated quiz templates and asynchronous task queues (Celery) to reduce API latency.
4.	Security	Standard API key environment variable configuration.	Integrate OAuth2 / Auth0 authentication, encrypted user data tables, and centralized secret management (AWS Secrets Manager).

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-User Authentication & Mobile-Responsive UI	Month 1 – Month 2	Enables personal account management, secured history tracking, and mobile accessibility.
Phase 3	Automated Adaptive Spaced Repetition Notifications	Month 3 – Month 4	Sends automated email/push notifications alerting users exactly when retention drops below threshold review levels
Phase 4	Multi-Modal Learning Support (PDF & Flashcard Ingestion)	Month 5 – Month 6	Allows users to upload study notes/PDFs directly to generate personalized quiz decks automatically.